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A plant-based diet is feasible in patients with Crohn's disease

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SUMMARY

Background: Incorporating plant-based diets as a supplement to medical treatment may have a beneficial impact on patients with Crohn's disease, however, research with intervention studies is required. **Objective:** To investigate the feasibility of a plant-based diet intervention. Secondly, the purpose was to investigate whether such diet may reduce disease activity and enhance quality of life.

Materials and methods: This study was designed as a single arm feasibility study. Outpatients with Crohn's disease in biological therapy were guided over twelve weeks towards a dietary lifestyle change. **Outcome measures:** Feasibility concerning recruitment, retention rate and compliance. Secondary outcomes were measures of patient reported outcome questionnaires (PROMS). Paired t-tests were used to examine changes in CO₂ emissions, anthropology, biomarkers, and patient-reported data. Δ-values were used to investigate difference between dietary intake and requirements. Linear regression analyses examined the association between biomarkers and PROMS.

Results: In total, 15 participants completed the intervention with easy recruitment and a retention rate at 87.6%. A clinically positive tendency was seen towards improved symptom scores for disease (HBI; $p=0.028$ and IBDQ; $p=0.006$) but not for fatigue (IBD-F; $p=0.097$), although none of these were statistically significant. Adverse effects were decreased protein intake ($p=0.069$) and slightly reduced muscle mass. It remains unclear to what extent the intervention contributed to the improved self-reported effects although perception of disease activity was improved.

Conclusion: This study demonstrates that it is possible to retain patients following a plant-based diet. However, the dietary change required ongoing dietetic support with a focus on anti-inflammatory agents and the still unattainable protein requirements.

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1. Introduction

Crohn's disease (CD) is an inflammatory bowel disease characterized by chronic transmural inflammation in the gastrointestinal tract. The disease follows a recurrent pattern of relapse and remission, with symptoms such as diarrhea, fever, abdominal pain, anemia, and fatigue [1]. CD occurs in genetically susceptible individuals, where environmental triggers [2]; an overactive immune response, and changes in gut microbiota contribute to chronic

inflammation and impaired intestinal barrier function [3,4]. Current medical therapies target the immune system and/or specific pro-inflammatory cytokines to achieve mucosal healing [5]. However, long-term remission is challenging, with 40% of CD patients experiencing relapse within a year [6]. Therefore, there is a need to identify complementary prevention and treatment strategies [7].

It is well-established that the microbiota in individuals with CD is less diverse and rich in species [8]. A traditional Western diet low in dietary fiber and high in animal protein and fats as well as highly processed foods worsens this condition [9,10]. Plant-based diets are rich in a range of potential anti-inflammatory components such as polyphenols found abundantly in various fruits and vegetables

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[9,10], and polyphenols have been shown to interact with different enzymes and cell receptors, exerting immune regulatory effects. Furthermore, plant-based diets may provide a diverse range of dietary fibers that enhance the gut microbiota's function and composition [11–13]. Despite this, a recent study found that patients often avoided specific foods after CD diagnosis and that foods most frequently avoided included spiced foods, seasoned foods, vegetables, legumes, and whole grain bread, which are known to be rich in anti-inflammatory agents [14]. Thus, whether a plant-based diet is viable for individuals with CD remains uncertain. Bloating which may be interpreted as symptoms by the individual, the presence of strictures, or personal dietary habits or concerns may result in the exclusion of food items crucial in a plant-based diet [11,15]. Furthermore, the transition to a strictly plant-based diet induces potential risks of nutrient deficiencies, especially vitamin B12 [16].

1.1. Objective

The objective of this study was to investigate recruitment, retention rate, and adherence to a 12 weeks plant-based diet intervention as complementary to medical treatment, and to examine any possible adverse perceived symptoms. Additionally, the study aimed to investigate whether a plant-based diet within 12 weeks may decrease disease activity and improve quality of life for individuals with CD.

2. Materials and methods

2.1. Study design and setting

This study was designed as a feasibility study without a control group or randomization process.

2.2. Participants

The study invited 15 patients aged ≥ 18 years (50.7 ± 15.4), diagnosed with CD for at least six months, and treated with biological medicine for at least three months. Furthermore, the disease had to be in remission or have maximum moderate activity as determined by the Crohn's Disease Activity Index (CDAI) [17]. A maximum variation sample was sought regarding age, sex, years of biological treatment, and education.

Patients who had been hospitalized due to exacerbation and patients who had changes to medication including steroid treatment within the past three months were excluded. An exclusion criterion for the study has therefore been patients who has experienced a CDAI score above 150, indicating active or extremely severe disease [17].

2.3. Inclusion

Participants were recruited through billboard notices placed in the treatment room where patients received biological treatment at the Department of Gastroenterology in a Danish University Hospital, as well as through snowball sampling. A physician screened patients who showed interest and offered them participation if eligible.

2.4. Intervention

Participants in the study received personalized guidance through weekly phone calls for a dietary lifestyle change. They were provided with information materials and recommendations for potential anti-inflammatory foods and sufficient protein intake

(Fig. 1). They also received free evening meals from a meal delivery company for 12 weeks, with a variety of options to choose from. Participants were guided to eat foods such as legumes, seeds, whole grains, vegetables, fruits, and berries, and to focus on the mono-unsaturated fatty acids from sources like nuts, seeds, soybeans, and olives [18]. Contacts were primarily undertaken by the full-time research assistant who is a dietitian with extensive experience with living with, as well as guiding towards a plant-based diet. The participants also had the opportunity to contact the project leader, a professor in applied nutrition, who had open access to the study gastroenterologist for any medical question posed. The participants received free evening meals from a meal delivery company for themselves and their families during the 12-week intervention, where they could choose from a weekly assortment of nine different meal options in a four-week rotation. The provided meals were plant-based, but participants could include fish and seafood twice weekly to ensure sufficient omega-3 fatty acids and protein intake. The meals were tailored in collaboration with the delivery company with a focus on food items with a high content of phenols as well as a solid intake of protein at approximately 25 g per serving. Furthermore, we elevated the estimated protein requirements to ensure that amino acid requirements were met in this 12-week intervention. Therefore, accounting for a generally lower digestibility and less balanced amino acid composition of plant protein, we used a pragmatic protein target of 1.1 g/kg/day instead of the target of 0.8 g/kg/day recommended for carnivore/mixed diets by the Danish Health Authorities [19].

Participants received weekly evening meals that only required heating in the oven or microwave, and they could contact the supplier for special dietary requests. Participants could thus choose between seven different plant-based options every week, and two dinner options with fish if desired. If there were items on the week menu that they did not like, they could exclude those, or if some were preferred more, they could choose as many of these meals as they liked, for instance, they could choose a lentil chili dish five times in a week if they wanted.

Through semi-structured interviews, we found that it was important to provide recipes and shopping guides [20]. Based on this, a recipe book for breakfast, lunch, and snacks, as well as a supermarket guide was created. The guide emphasized eating processed foods with primarily omega 3 unsaturated fatty acids from sources of vegetables, nuts and seeds, fiber-rich products, whole grains, and products with a moderate salt content. Protein intake was also prioritized, with recommendations for legumes, tofu, quinoa, nuts etc. [21,22]. Participants were encouraged to take vitamin B12 and D supplements. Weekly phone calls or email correspondence provided dietetic advice from the project assistant (dietician, M. Sci).

2.5. Feasibility

Feasibility was assessed and reported with regard to inclusion, retention and compliance [23,24]. Compliance to the intervention was checked by the dietary registration as well as unsystematically during the weekly phone calls and e-mail correspondence. Additionally, a project employee followed up on the participants at baseline (W0), week four (W4) and week 12 (W12) after inclusion.

2.6. Anthropometric and blood pressure measurements

Anthropometric indices including height, weight, body mass index (BMI), fat mass, and muscle mass were measured at W0, W4, and W12. At W0 height was measured using a Seca stadiometer, and participants were weighed on a Seca scale while they wore light clothing and were barefoot. Body composition was measured



Fig. 1. Poster of the principles of the diet.

by bioimpedance using an InBody S10. Prior to the measurement, the participants had refrained from eating or drinking for 4-h and had used the restroom within 30 min. Additionally, the participants were required to lie down and relax for 10 min, barefoot and without any metal objects such as belt and watch.

2.7. Laboratory analysis

Blood and stool samples were obtained before the start of the intervention, at W4, and W12. After a 4-h fast, our affiliated medical laboratory technologist collected the blood samples in Vacuette FC mix tube 3 ml for Fasting glucose (fPt-glucose), Vacuette K2 EDTA 4 ml for hemoglobinA1C, mean blood glucose (HbA1Ci), leucocytes, mean corpuscular thrombocyte volume (MCV), and in Vacuette Li-Hep 4 ml, albumin (alb), vitamin B12, carbamide, cholesterol, creatinine, C-reactive protein (CRP), D-vitamine (D-tot25), estimated glomerular filtration rate (eGFR2), ferritin (Fe), Fe saturation (Fe-sat), folic acid, fasting LDL Cholesterol (fPt-LDL), High-density lipoprotein cholesterol (HDL cholesterol), potassium (K), magnesium (Mg), sodium (Na), triglyceride (PfPt). Further, 4 × 0,5 ml was frozen at 80 °C (Vacuette Li-Heparin, 3000/min) in 10 min. Stool samples were sampled by the patient and frozen at home using a provided sampling and storage kit and brought to the hospital less than 24-h before the blood sampling time. The samples were processed by the laboratory by usual clinical practice. Stool samples were analyzed for fecal calprotectin (F-calprotectin) level by the method of Atellica CH, turbidimetry; fCAI turbo by Bühlmann [25]. The results were followingly retrieved through the patients' medical journal. Reference values were set at <200 mg/kg: no disease activity, 200–600 mg/kg: moderate disease activity and >600 mg/kg: severe disease activity [26–28]. CRP was analyzed by the method of Abbott, Alinity (immunoturbidimetric method, CRP 48) [29].

2.8. Patient-reported data

At baseline the participants completed a questionnaire including background information and questions about disease. At W0, W4, and W12, the Harvey-Bradshaw Index (HBI) [30], using the five variables general well-being, abdominal pain, abdominal mass, number of liquid stools, and systemic complications, was completed. Furthermore, the participants completed the Inflammatory Bowel Disease Questionnaire (IBDQ) [31]. The questionnaire measured self-perceived health-related quality of life (HRQoL). The questionnaire consisted of 32 questions that are categorized into four dimensions: bowel symptoms, systemic symptoms, emotional function, and social function [31]. The participants also completed the Inflammatory Bowel Disease-Fatigue Self-assessment Scale (IBD-F) which is a disease-specific scoring system to rate levels of fatigue [32–34]. The score does not necessarily correlate with disease activity [32,33], but the scale has demonstrated a strong correlation with the IBDQ and other fatigue assessment tools [34].

2.9. Dietary registration

Participants kept a three-day dietary registration at W0, at W4 and just before the end of the intervention, with a method often used for research as well in clinical practice in this setting [35]. The participants were asked to register intake one day in the weekend and two weekdays on pre-printed forms, within the week before the three sampling timepoints. At the appointments the research assistant (trained dietician) and the participant went through the forms together, to minimize the risk of over- and under reporting best possibly [36]. The participants' energy- and protein intake as well as carbon footprint in terms of carbon dioxide equivalents

(CO₂e) was calculated based on their dietary records using the program “DankostPro” [37].

2.10. Statistical analysis

Statistical analyses were conducted using Stata (version 18.0. Stata Corp, College Station, TX, USA). For variables that were normally distributed, mean, and standard deviation (SD) was calculated, while for variables that were not normally distributed, median and range [minimum and maximum] were calculated. Shapiro-Wilk test was performed to test for normal distribution. Additionally, paired t-tests were performed to compare the changes in dietary intakes, CO₂ emissions, CRP levels, and patients' f-calprotectin levels over the project period for normally distributed variables, while Wilcoxon matched-pairs signed-rank tests were used for variables that were not normally distributed. In addition, paired T-test were calculated to examine the difference between W0 and W4, W0 and W12 as well as W4 and W12 for each participant. In the comparison of Δ -values for calorie and protein intake, these were compared to participants' energy and protein needs to examine the extent to which their individual requirements were met (Table 3). The bioanalytical measurements (CRP and F-calprotectin) were compared to the self-reported data (IBDQ, HBI, and IBD-F) using linear regression analysis. One patient could not be measured at W12, and therefore paired t-test, the Wilcoxon matched-pairs signed-rank and the linear regression analysis were performed with $n = 14$. A significance level of $P < 0.05$ was chosen and a coefficient and 95% confidence intervals were calculated.

2.11. Ethical considerations

Participation in the intervention was voluntary, and participants were required to sign a written declaration of consent prior to commencement, which included granting access to medical records. The study was conducted following the guidelines outlined in the Helsinki Declaration of 2002, which allowed participants to withdraw their consent at any time. The participants received free, plant-based dinners for themselves and their families, with the option to add fish and seafood twice per week.

3. Results

3.1. Participant characteristics

The 15 participants, eight females (53.3%), who completed the study were fairly age distributed for ≥ 18 -year-olds with the diagnosis (50.7 ± 15.4) (Table 1). Educational level was common for the general Danish population, however only nine participants worked full time. Only one participant lived alone and eight of the participants had children living at home (Table 1). Most participants had their diagnosis between age 17 and 40 years (53.3%). Two of the participants were smokers and only one had been hospitalized within the past year due to flare of CD. Seven participants had their disease localized to the L1 ileal segment (46.7%), five had it localized to the L2 colonic (33.3%), while three patients had their disease localized to the L3 ileocolonic (20.0%) (Table 1). The most common medication was Infliximab (73.3%), with an average duration of use for 68.2 ± 49.1 months.

3.2. Primary outcome: feasibility of the study

3.2.1. Recruitment and retention

The study recruited the pre-defined 15 participants within two weeks. Already during the second week of the study, two women had lack of compliance due to dislike of the plant-based diet.

Table 1

Baseline characteristics of all study participants at entry and characteristics about the disease ($n = 15$).

Variable	N (%) or mean (SD)
Age (y):	50.7 $\pm$ 14.4
18–40	5 (33.3)
41–60	5 (33.3)
>60	5 (33.3)
Education:	
No	2 (13.3)
Vocational education	3 (20.0)
Short education	4 (26.7)
Medium education	3 (20.0)
Long education	3 (20.0)
Employment situation:	
Full time job	9 (60.0)
Retired	2 (13.3)
Early retirement	4 (26.7)
House composition	
Marital status:	
Single	1 (6.7)
Cohabiting	14 (93.3)
Members of the household:	
1	1 (6.7)
≥ 2	14 (93.3)
Children living at home: yes	7 (46.7)
Lifestyle	
Smoking/other nicotine products: yes	2 (13.3)
Drinking alcohol: yes	13 (86.7)
Number of alcohol units per week:	4.2 $\pm$ 3.0
1–5	9 (69.2)
6–10	4 (30.8)
Comorbidities: yes	10 (66.7)

Therefore, the participants were completely withdrawn from the project and are not included in the analysis (Fig. 2). Instead, two new participants were recruited by asking the physicians and nurses in the outpatient clinic if they could find two patients who fulfilled the criteria. These were thus included approximately three weeks into the project. Additionally, one male participant was involuntarily lost to follow-up at W12 due to hospitalization, which prevented us from obtaining his data at the end of the intervention. However, the measures were still included in the data for W0 and W4 analysis. The participant retention for the project was 86.7%.

3.2.2. Adherence to the intervention

The unsystematic compliance check revealed no trouble in adherence to a plant-based diet for breakfast, lunch, and snacks. Three participants struggled to stick to the diet, opting for white bread and toppings instead of healthier options as suggested. They also had difficulty covering protein requirements, preferring almond or oat beverages and sugary yogurt over soy products. After a few weeks, many participants found the pre-made dinners monotonous and wanted guidance on adding fresh sides. They received instructions on alternative dietary options and received instructions on the intake of digestible and indigestible carbohydrates, as well as guidance on alternative dietary options based on macronutrient composition.

3.2.3. Adverse effects

No adverse events or effects were reported. Only one hospitalization was seen due to flare-up during the intervention period, which left him out of W12 analysis. Checking this patient's food records, no reason was found to blame the intervention, as the patient reported sufficient change to a plant-based diet and sufficient food intake which were registered up to the days of hospital admission and may rather be due to the common loss of effect by Adalimumab [38].

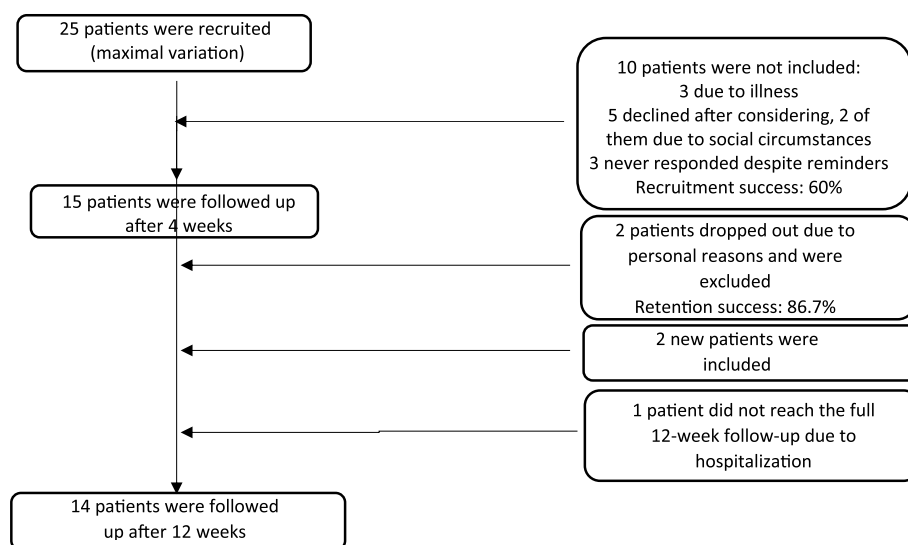


Fig. 2. Flowchart of study progress.

3.3. Secondary outcomes

3.3.1. Carbon footprint

The study demonstrated an association between CO₂ emissions and the participants' dietary intake. The three-days dietary records, completed by the participants, showed that all the participants consumed meals that included a lower carbon footprint at W4 and W12 than at W0 (Table 2). The calculations showed that the carbon footprint decreased from W0 to W4 ($p < 0.001$) and from W0 to W12 ($p < 0.001$). There was no difference between W4 and W12 ($p = 0.883$) in carbon footprint (Table 3). The calculated changes in CO₂ emissions suggested that during the intervention period, the participants consumed food that had a lower climate impact compared to before the project commenced.

3.3.2. Food intake

There was a significant decrease in calorie intake from W0 to W4 ($p = 0.015$). When comparing W4 to W12, there was no significant change in calorie intake ($p = 0.534$), and the same applied when comparing W0 to W12 ($p = 0.157$). The calculations of dietary records indicated that four of the participants experienced an increase in calorie intake from W0 to W12, while ten participants consumed fewer calories ($p = 0.157$) (Table 3). This leveled off

somewhat in the coming weeks (Table 2). The protein intake also decreased when comparing W0 to W4 ($p = 0.002$) and W0 to W12 ($p = 0.069$), with a slight increase between W4 and W12 ($p = 0.248$) (Table 3). The averages from the three-day dietary recall indicate that ten participants consumed less protein during the intervention period compared to W0 values.

3.3.3. Achieved dietary intake compared to requirements

The comparison of participants' energy requirements of 20 kcal/kg to intake of calories and proteins showed that most participants had at minimum met their caloric needs at W0, W4, and W12 (Table 4). When comparing protein intake with the requirement, it was observed that seven participants at W4 and five participants at W12 only achieved 50–74% of their protein needs.

3.3.4. Body weight and body composition

The participants remained relatively stable over the 12 weeks of the intervention. The change in participants' weight by an average of 2.21 kg from W0 to W12 was not statistically significant ($p = 0.541$). Notably, however, one participant went from being classified as overweight based on their W0 BMI to being categorized as having a normal BMI by W12.

Table 2

Change over time in CO₂ emissions, calorie, and protein intake.

Variable	W0 – W4	W4 – W12	W0 – W12
	N (%) or mean $\pm$ SD N = 15	N (%) or mean $\pm$ SD N = 14	N (%) or mean $\pm$ SD N = 14
CO₂ average (kg CO₂/unit)	–2.6 $\pm$ 1.3	0.0 $\pm$ 1.1	–2.6 $\pm$ 1.1
Unchanged	0 (0.0)	0 (0.0)	0 (0.0)
Increased	1 (6.7)	10 (71.4)	0 (0.0)
Decreased	14 (93.3)	4 (28.6)	14 (100.0)
Calorie intake^a (Kcal %)	–14.2 $\pm$ 21.1	–4.3 $\pm$ 28.4	–9.8 $\pm$ 23.1
Unchanged	0 (0.0)	0 (0.0)	0 (0.0)
Increased	3 (20.0)	8 (57.1)	4 (28.6)
Decreased	12 (80.0)	6 (42.9)	10 (71.4)
Protein intake^a (g %)	–22.7 $\pm$ 23.7	8.6 $\pm$ 26.3	–14.6 $\pm$ 23.1
Unchanged	0 (0.0)	0 (0.0)	0 (0.0)
Increased	2 (13.3)	7 (50.0)	4 (28.6)
Decreased	13 (86.7)	7 (50.0)	10 (71.4)

^a Calorie and protein intake in relation to requirements, between W0 to W4, W4 to W12 and W0 to W12.

Table 3

Change in body weight, body composition, CO₂, calorie- and protein intake, CRP, F-calprotectin, and self-reported data from start of the intervention to week 4 and from week 4 to week 12 of the intervention as well as week 0 to week 12.

Variabel	Week 0-Week 4°			Week 4-week 12°			Week 0-week 12°		
	Week 0 mean ± SD	Week 4 mean ± SD	p-value	Week 4 mean ± SD	Week 12 mean ± SD	p-value	Week 0 mean ± SD	Week 12 mean ± SD	p-value
Calorie intake, Kcal (%)	1885.2 ± 413.7	1645.5 ± 384.5	0.015*	1683.9 ± 571.9	1697.9 ± 362.5	0.534	1821.6 ± 522.5	1697.9 ± 362.5	0.157
Protein intake, g (%)	73.4 ± 16.9	56.3 ± 12.6	0.002*	55.4 ± 12.6	61.7 ± 12.5	0.248	72.8 ± 17.4	61.7 ± 12.5	0.069
Weight (kg)	81.9 ± 14.9	80.7 ± 14.4	0.122	81.2 ± 14.8	79.7 ± 14.8	0.0051	82.0 ± 15.4	79.7 ± 14.8	0.0050*
BMI (kg/m ²)	26.9 ± 4.9	26.5 ± 4.7	0.115	26.9 ± 4.6	26.4 ± 4.6	0.006*	27.2 ± 5.0	26.4 ± 4.5	0.006*
Muscle mass (%)	37.5 ± 5.6	37.6 ± 5.4	0.801	37.2 ± 5.5	37.4 ± 5.7	0.418	37.3 ± 5.8	37.4 ± 5.7	0.724
Muscle mass (kg)	30.5 ± 6.2	30.2 ± 6.0	0.041*	30.1 ± 6.2	29.6 ± 6.2	0.005*	30.4 ± 6.4	29.6 ± 6.2	0.001*
Fat mass (%)	32.2 ± 9.6	31.7 ± 9.5	0.300	32.4 ± 9.4	32.2 ± 9.7	0.469	32.7 ± 9.7	32.2 ± 9.7	0.244
Fat mass (kg)	26.8 ± 10.5	26.0 ± 10.2	0.150	26.7 ± 10.2	26.1 ± 10.2	0.060	27.3 ± 10.7	26.1 ± 10.2	0.030*
CO ₂ average (kg CO ₂ /unit)	4.2 ± 2.0	1.6 ± 0.9	<0.0001*	1.5 ± 0.9	1.6 ± 0.4	0.883	4.1 ± 1.1	1.6 ± 0.4	<0.0001*
CRP (mg/l)	6.4 ± 11.6	8.0 ± 13.1	0.573	7.9 ± 13.6	8.0 ± 14.0	0.993	6.5 ± 12.0	8.0 ± 14.0	0.659
F-calprotectin (mg/kg)	442.4 ± 658.0	401.5 ± 619.0	0.238	366.7 ± 627.0	208.0 ± 326.0	0.315	413.4 ± 672.7	207.6 ± 395.7	0.212
IBD-F total score	10.4 ± 5.0	8.7 ± 3.7	0.156	8.8 ± 3.8	8.4 ± 4.5	0.716	10.8 ± 5.0	8.4 ± 4.5	0.097
HBI total score	3.2 ± 2.0	2.7 ± 1.7	0.301	2.3 ± 1.9	2.4 ± 2.6	0.828	3.1 ± 2.0	2.4 ± 2.6	0.028*
IBDQ total score	173.3 ± 26.3	184.7 ± 20.5	0.045*	187 ± 19.0	191.4 ± 25.0	0.400	173.0 ± 27.3	191.4 ± 25.0	0.006*

*p < 0.05.

° n = 15 at week 0 and week 4 but n = 14 at week 12.

BMI = Body Mass Index. CRP=C-reactive Protein. F-calprotectin = Fecal calprotectin. IBD-F = Inflammatory Bowel Disease Fatigue self-assessment scale. HBI = Harvey Bradshaw Index. IBDQ = Inflammatory Bowel Disease Questionnaire.

Table 4

Achieved individual dietary intake of calories and protein relative to individual requirements (%) and number of patients within five categories of level of achievement.

	Week 0 N = 15	Week 4 N = 15	Week 12 N = 14
Achieved relative to caloric needs (%)	116.5 ± 23.7	102.3 ± 25.0	105.7 ± 23.3
<25%	0 (0.0)	0 (0.0)	0 (0.0)
25–49%	0 (0.0)	1 (6.7)	0 (0.0)
50–74%	1 (6.7)	0 (0.0)	1 (7.1)
75–100%	3 (20.0)	5 (33.3)	4 (28.6)
>100%	11 (73.3)	9 (60.0)	9 (64.3)
Achieved relative to protein needs (%)	99.4 ± 23.0	76.6 ± 19.0	84.8 ± 19.0
<25%	0 (0.0)	0 (0.0)	0 (0.0)
25–49%	0 (0.0)	1 (6.7)	0 (0.0)
50–74%	2 (13.3)	7 (46.7)	5 (35.7)
75–100%	6 (40.0)	5 (33.3)	5 (35.7)
>100%	7 (46.7)	2 (13.3)	4 (28.6)

Neither changes in participants' muscle mass (−0.15%), fat mass (+0.01%), nor BMI (−0.55 kg/m²) W0 to W12 were statistically significant.

The participants' muscle mass on average, changed by −0.3 kg from W0 to W4 (p = 0.041) and by −0.5 from W4 to W12 (p = 0.005) and from W0 to W12, decreased by 0.8 kg (p = 0.001) (Table 3). One participant remained stable, four participants gained muscle mass, while ten participants lost muscle mass (kg).

3.3.5. Self-reported data

Self-reported data formed the participants' subjective experiences of how the intervention affected them (Table 3). Overall, the study indicated that the total score of the IBD-F worsened among eight participants from the baseline assessment to the final measurement. When comparing W0 to W12, two participants remained stable in their fatigue symptoms, while four patients experienced an improvement, and eight participants reported a worsening of fatigue. Overall, there were no significant changes in the self-reported score IBD-F (Table 3).

The HBI score showed an overall positive trend in participants' experiences of disease activity and the consumption of a plant-based diet from W0 to W12 (p = 0.028) (Table 3). Ten participants reported overall improvement from W0 to W12, while two experienced worsening and two reported no change.

The IBDQ that assesses participants' self-perceived HRQoL, improved overall from W0 to W12 (p = 0.006) (Table 3). Ten participants reported an improvement in HRQoL, while five participants experienced a worsening from W0 to W4. Comparing W0 to W12, 12 participants experienced improvement in quality of life measured by HRQoL, while two participants found a worsening.

3.3.6. F-calprotectin

No significant change of F-calprotectin levels from W0 to W4 and W12 were observed (Table 3). The stool samples showed a variation in development between participants. The F-calprotectin levels remained stable among four patients, while four participants experienced an increase, and the other six participants had a decrease in levels.

3.3.7. CRP

No statistically significant change was obtained for CRP for the whole cohort (Table 3). However, eight participants experienced a decrease in CRP from W0 to W12, while three participants experienced an increase in levels, and three unchanged CRP levels.

3.3.8. Associations between self-reported data and bioanalytical measurements

The associations between CRP levels and F-calprotectin with the self-reported IBDQ, HBI, and IBD-F questionnaires were examined

Table 5
Associations between the PROMS IBDQ, HBI and IBD-F and bioanalytical measurements.

	Coefficient (week 0)	p-value (week 0)	Confidence interval (week 0)	Coefficient (week 12)	p-value (week 12)	Confidence interval (week 12)
IBDQ						
CRP	−0.20	0.752	−1.56–1.16	−1.224	0.010*	−2.10 – −0.36
F-calprotectin (mg/kg)	−0.00	0.666	−0.03–0.02	−0.01	0.795	−0.05–0.04
HBI						
CRP (mg/l)	0.05	0.26	−0.04–0.15	0.15	0.001*	0.071–0.22
F-calprotectin (mg/kg)	−0.00	0.843	−0.00	−0.00	0.921	−0.00–0.00
IBD-F						
CRP (mg/l)	−0.09	0.434	−0.35–0.16	0.05	0.587	−0.15–0.26
F-calprotectin (mg/kg)	0.00	0.975	−0.00–0.00	0.00	0.235	−0.00–0.00

*p < 0.05.

by linear regression analyses (Table 5). These showed a negative correlation between CRP levels and IBDQ at W12 ($p = 0.010$). Additionally, a positive correlation was observed between CRP levels and the reporting of the HBI questionnaire ($p = 0.001$). The remaining calculations did not show any correlation between levels of CRP or F-calprotectin and the self-reported questionnaires.

4. Discussion

4.1. Recruitment

The intervention was feasible in recruiting patients. Our target patients showed high interest in improving their well-being and minimizing disease symptoms. When two participants dropped out, the project team easily recruited two new participants, confirming the feasibility of participant recruitment. It is relevant to discuss whether recruiting participants with severe disease activity would have been as easy as recruiting those with mild to moderate disease activity. However, a study found that recruiting patients with mild disease activity was more challenging compared to those with severe disease activity [39]. Additionally, the study mentioned the difficulty in recruiting patients with inflammatory bowel disease from the local community, requiring web-based community sites for broader representation [39]. In the present study, we managed to recruit a diverse group of participants within the local area. The patients were affiliated with the Danish hospital where they received biological treatment.

4.2. Retention

The daily pre-made dinners given to the participants aimed to increase retention but turned out to requiring more motivational calls than first anticipated. While the participants appreciated the convenience, they grew tired of the repetitive food, which affected their motivation to stay on the intervention diet. The retention rate in the intervention was 86.7%, with one participant unable to complete the final measurements due to hospitalization. Two participants dropped out, partly due to difficulties with the meals. In a less invasive lifestyle intervention where participants with CD had to eat vegetables, fish, and light meat for six months, the drop-out was 3/29 within eight weeks, and in total 83% of the participants completed the study [40]. Overall, participant retention was high in this intervention. Despite some dissatisfaction with the pre-made meals, it is worth considering if it contributed to the high retention. The meal provider was responsive to requests, and the project team encouraged participants to supplement the meals for more excitement.

4.3. Adherence

The participants were highly motivated to engage in the intervention prior to its initiation, and overall, they did not encounter

difficulties in adhering to the plant-based diet. However, it became obvious that maintaining strong motivation towards the dietary principles was challenging during periods of social gatherings. Another study, examining the relationship between social norms and their influence on dietary intake, also showed that social contexts have an impact on what people eat, and the study revealed that conforming to or deviating from social norms was associated with judgments [41]. Additionally, it was observed that social norms could influence food choices [41]. To assist participants during these occasions, the project assistant provided ongoing guidance on how to avoid deviating from the dietary principles of the project.

4.4. Food intake

The protein intake changed significantly from W0 to W12. Seven participants achieved 50–74% of their protein needs in W4. A study showed that in plant-based diets, the protein intake was comparatively lower than other diet types [42], but even though the amount of protein is relevant to consider, it may also be important to enhance a diversity in the sources of plant-based proteins and account for the varying amino acid content and digestibility to match requirements and ensure proper amino acid composition [22,43]. Therefore, the focus was on the nine essential amino acids for nutritional intake, and the supplier was given instructions on the composition of protein-rich sources from nuts and seeds, legumes, tofu, vegetables, and whole grain products. Furthermore, the delivered meals contained 25 g of protein and fish twice a week, raising the question whether the protein intake may have been lower without the delivered food.

Some participants were unfamiliar with plant-based foods, so substitutes were discussed to increase protein intake. The study did not include other animal meat sources besides fish and seafood, however analysis of compliance to this part was not investigated and the participants food records were used as the value of truth in the study. The protein intake slightly improved from W4 to W12 due to the increased focus on protein, emphasizing nutrition-rich sources in a plant-based diet. So, ensuring enough protein intake should be a key focus in similar interventions.

4.5. Body composition

One of the secondary objectives were to evaluate the effect of the diet based on measurements of body composition. The study's bioimpedance measurements indicated that most participants lost weight during the study. When people eat plant-based foods, some tend to consume fewer calories because these foods generally have lower energy density compared to animal products, which could be attributed to higher consumption of dietary fiber, energy density and utilization, and plant proteins while reducing their intake of saturated fat and animal protein [44]. Unfortunately, this also led to a slight but from a clinical aspect insignificant reduction in muscle

mass, which may be due to the reduced protein intake. A potential concern is also the slightly increased fatigue, despite there being no statistical association. One may worry that a longer term, unsupervised dietary change to a plant-based nutrition, may decrease protein intake further, with possible negative effects to muscle mass. Other studies have found that exercise reduce fatigue in patients with CD [45], probably due to increase muscle mass and appetite as well as decrease in fatigue. In the present study the importance of exercise was superficially addressed but did not specifically encourage daily physical activity.

4.6. Self-reported data

There were a few missing data in the self-reported questionnaires. A few participants found certain questions in the IBD-F and IBDQ questionnaires challenging to answer. Some participants also suggested that the excessive number of questions, indicating a need to consider reducing the questionnaire quantity. Another relevant questionnaire could be the shorter version of the IBDQ with ten questions instead of 32 (SIBDQ) [46]. The questionnaires used in the study were pre-defined and well-tested, specifically tailored to the patient group. These questionnaires ensured the validity of the study by measuring relevant things to investigate. For making sure patients remained in disease remission or moderate disease, measures of CRP and F-calprotectin were included to support the PROMS. When suffering from CD, the level of calprotectin in feces is expected to increase, and would further increase on the onset of disease activity [47]. Therefore, it was expected that the level of F-calprotectin would also indicate the participants' disease activity and thus the individual's degree of inflammation [28]. The values of PROMS were reflected in increased CRP, however not for IBD-fatigue. In contrast, there was an association with F-calprotectin, which seem aligned with HBI and IBDQ. However, all measures only differed slightly and were not pronounced at any of the times of measuring. The association between PROMS and inflammation markers should be tested in the same population with a wider range of measures.

4.7. Ethical considerations

Participants received complimentary dinners for themselves and their families during the intervention, and although this was not the intent, it may be argued that this have influenced the recruitment. However, the advantage of being part of the intervention in the form of free meals is not assumed to be the decisive motivating factor, as it was expressed later in the intervention period that participants became tired of the provided dinners and started making or supplementing their own dinners based on the dietary principles.

4.8. Adverse effects

No adverse effects were reported.

4.9. Strengths and limitations of the study

This study involved 15 participants who were asked to share their individual perceptions. It is important to note that these perceptions are not intended to be generalizable to all CD patients or their experiences. Additionally, all participants in this study were patients who were almost in remission due to biological therapy and were living in a well-developed country with ample access to healthcare, medication, and food. It is therefore not appropriate to generalize the findings of this study to all countries.

5. Conclusion

This feasibility study demonstrates that it is possible to retain patients in following a plant-based diet. However, the dietary change required ongoing dietetic support with a focus on anti-inflammatory agents and the still unattainable protein requirements. Unclear effects were shown for secondary outcomes.

Ethical approval and concerns

The project was approved by the Scientific Ethics Committee for the North Jutland Region on December 2, 2022 (N-20220049), in accordance with the Act on the Scientific Treatment of Health Science Research Projects and Health Data Science Research Projects, and subsequent committee legislation.

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Authors' contributions

CL, MH, and HHR conceptualized, and MH, HHR, ZC, SKØ and HNL planned the study. LA and MH collected the data. LA and SKØ analyzed the data with contributions from SM. LA wrote the first draft assisted by MH. All authors reviewed and commented on subsequent drafts of the manuscript. All authors agreed to the final edition.

Availability of data and materials

Data and material with no personal information will be made available at request after all detail analysis have been made and published.

Declaration of AI and AI-assisted technologies in the writing process

No AI or AI-assisted technologies were used.

Declaration of competing interest

None.

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